

INSTITUTE OF ACTUARIES OF INDIA

Subject SA4 - Pensions and Other Benefits

November 2024 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1**i. Company:****Pros:**

- Potential cost saving if the commutation factors are computed using optimistic basis.
- Reduced mortality risk, interest rate risk for future years.
- Members may appreciate such options as different people have different circumstances and financial needs
- May help in reducing tax outgo

Cons:

- Potential cashflow strain and liquidity risk if many members accept the commutation
- Risk of anti selection as members in bad health likely to opt for this option
- Once implemented, the company may need to continue this option in all future years to maintain equity across all members.
- Younger members may demand at least a portion of benefit to be commuted.
- Option take up rate may be very low if the members perceive the terms of commutation to be not attractive.

Trustees :**Pros:**

- Reduced future administration costs as lower number of members need to be serviced
- Lower amount of assets and liabilities to be managed

Cons:

- Increase in administration costs and efforts when member opts for commutation
- Need effective communication to reach out to all the members
- Need educative material and SOP in case members seek advise
- Need sound actuarial advise to ensure equitable distribution of assets to all members while ensuring fair commutation factors.
- If scheme is under funded and the commutation value is paid out of the fund, could lead to worsening of funding position.

Members:**Pros:**

- Additional benefit option made available.
- Members in need to immediate lumpsum may benefit
- Financially literate members may be able to generate better returns

Cons:

- Possibility of inefficient decision as members forego future income for a lumpsum.
- The lumpsum can be mismanaged if members are not savvy enough.

(9)

ii. Data required for preparing the Commutation table are -

- Mortality table adopted by the trustees
- Past mortality experience of the scheme to allow for any scheme specific adjustments

- Interest rate curve
- Overall liability and asset position of the trust

Method of estimating the table -

- The commutation factors are simply single life annuity factors
- Expected future pension cashflows would be projected based on the assumed mortality rates after allowing for any future improvements or scheme specific adjustments.
- On discounting these cashflows we can estimate the present value of pension outgo at the given age.
- Such calculation needs to be repeated for each age
- the commutation factors may be expressed as lumpsum for each rupee of pension or each INR 1000 of pension.
- Value of factors will vary significantly at each age as the mortality increases steeply at advanced ages. Hence need to specify whether and how the factors can be interpolated.

Other considerations

- Need to consider the funding level of the trust. Any lumpsum outgo will affect the security of benefits for the continuing members.
- Need to consider the expected reduction in interest rates in India in the future years. This can be accounted for by reducing the discount rate assumption for every future year by 10 / 20 basis points upto a certain floor.
- Reducing discount rate assumption will iron out the risk of bond yields being high at the point of time when commutation table is formed for the members.
- Trustees may provide an input to change the assumptions and actuarial factors to reduce the commutation lumpsum in case the scheme is severely underfunded.
- Need to strike balance between favourable rates (to encourage take up of option) and sustainable rates (that can be offered to members in future year without affecting the scheme adversely)
- The strength of the assumptions depends on the objective of the company to encourage take up rate and company's ability to provide the requisite cashflows without affecting the funding level
- Trustees would want the best terms for members taking up the option, without compromising on the security of remaining members.
- Alternative scenarios with explanation of financial implications of each scenario need to be provided to the company /trustees.

(6)

iii. The member could have received a slightly higher pension from market if:

- The Company may have offered liberal commutation rates to encourage take up
- The Company may be using lighter mortality assumption than the one used by insurer leading to higher lumpsum in lieu of foregone pension.
- There may have been some gap between receipt of the commutation amount and purchase of annuity. Member could get higher pension benefit at age increases.
- Insurer may have offered beneficial terms to acquire market share.
- Member may have invested the assets and received windfall gains on the assets such that corpus available for purchase of annuity increased quickly

- The Company would have computed commutation lumpsum at a lower Interest rate and by the time this amount was used to buy annuity, interest rates might have gone up
- Health of the member might have deteriorated which would lead to better pension benefits with reduced life expectancy

(4)

iv. Financial feasibility of increasing pension benefits for other pensioners -

- Need to evaluate the value of defined benefit obligation on current set of rule and pension in payment
- Need to evaluate the value of defined benefit obligation on proposed set of rule and proposed pension in payment basis the same set of actuarial assumptions
- The increase in valuation of defined benefit obligation is the impact due to the proposed benefit enhancement.
- Need to confirm with the management regarding allowance required for future increases after implementing the proposed increase.
- Need to compare the assets available under both the scenarios.
- If the assets are adequate or if the deficit can be easily funded by the sponsor the benefit increments may be feasible.
- In case the deficit after implementation of the enhancements is unacceptable, a contribution plan needs to be prepared

Other practical considerations

- Verify the validity of the claim that member was able to get higher amount through annuities and whether such trend is visible for all members at all ages.
- Assess whether the future pensions will be eroded by inflation if this increase is not provided, i.e. estimate the future inflation to arrive at the % increase.
- Need to check if purchasing annuities for all the members is more feasible than awarding 10% increase.
- If such increments are awarded, the members may demand similar increases in future years. Financial impact of such future increases to be considered.
- The active employees can also demand an increase. And financial impact of increase for active members.
- Company can try to offer other alternative benefits like additional pension for members crossing 70 years of age or a dependents benefit.
- Need to check with company regarding willingness and capacity to meet the additional costs in such a way that security of existing benefits is not impacted adversely.
- whether the higher pension offered by insurance company is a one off case or generally higher pensions are offered by all insurance companies.
- Company / Trustees may want to re-look at the asset mix/ investment strategy.

(8)

v. Impact on IND AS 19 expense reporting due to benefit enhancement

- Under IND AS 19, the consolidated expense comprises of cost recognized in P&L and remeasurements recognized in OCI
- Employee benefit expense comprise of service cost (current service cost + past service cost) and net interest cost
- Remeasurements comprise of gains/losses due to change in demographic assumptions, gains/losses due to change in financial assumptions, gains/losses

due to participant experience, gains/losses due to difference in actual and expected returns on assets.

- Impact of any changes to benefit structure is immediately recognized as past service cost in P&L expense under IND AS 19
- The past service cost will be estimated as difference in obligation computed on old and new set of scheme rules as on the date of revision in rules.
- The past service cost needs to allow for interest cost for half the year that elapsed after the change in rules.
- The service cost projected on the opening date will need to be restated due to the impact of higher accrual. The impact needs to be computed for period for which new rules are applicable (Oct-Mar period)
- The option for commutation leads to “settlement accounting” of benefit.
- The gains or losses on settlement are recognized immediately in P&L.
- The amount of gains or losses on settlement will be computed based on the difference in the amount of obligation as on the date of settlement and amount actually paid on settlement.
- The interest cost will be same as that projected at the start of the year.
- The gains / losses due to change in demographic assumptions or financial assumptions will be computed as difference in value of obligation computed using basis as on the previous valuation date and the new basis adopted for the current valuation.

(10)

vi. Statutory pension scheme in India

EPS –

- Employees earning a base salary plus DA of Rs.15,000 or less are required to enroll in the scheme.
- The minimum monthly pension amount that the individual will receive is Rs.1,000.
- Provides a fixed income after retirement at the age of 58 years or after early retirement at 50 years
- Maximum pensionable salary is INR 15,000
- Amount of pension = pensionable salary * service /70
- Provides reduced pension to the member’s family in case of death of the member before or after the pensionable service period .
- The employer and employee contribute 12% of the employee's basic salary and DA towards the EPF scheme. Of this 8.33% of INR 15,000 is allocated towards EPS and balance towards EPF. (there are no employee contributions)

Voluntary benefit

National Pension System (NPS):

- Corporate NPS is available for Public and Private sector companies and is offered as an employee benefit along with other statutory benefits
- NPS is regulated by PFRDA, so there’s no fiduciary duty for employer
- Each employee is allocated an NPS account and the contributions are credited to this account.
- Employee makes all the investment choices and bears the risk.
- At retirement, 60% of lumpsum can be withdrawn as tax free and remaining 40% is required to purchase annuity pension from an insurer of choice.
- Withdrawal can be staggered and deferred as well until 75 years of age

- Employer's contribution of up to 10% of Salary (Basic + Dearness Allowance) up to 750,000 p.a. is deductible from taxable income under section 80CCD (2) over and above Rs.150, 000 limit u/s 80 CCE. In the 2024 Budget announcement, employer's contribution can go upto 14% of salary (Basic + Dearness Allowance) if the employee opts for New Tax regime.
- There is no maximum contribution specified in Corporate NPS and the above limits of 10% / 14% of salary are only with respect to the tax-exempt limits.

Other DC pension:

- Company can start and manage a DC scheme with rules set by the company like Superannuation scheme
- Employer contribution of upto 15% of basic salary is tax free
- At retirement, upto 1/3rd of the corpus can be withdrawn tax free and remaining 2/3rd is required to purchase annuity pension which is taxable
- Administratively cumbersome compared to adopting NPS

Other DB pension:

- Company can start and manage another DB scheme with benefits lower than currently promised.

(9)

[46 Marks]

Solution 2:

- i. The accounting standard categorizes employee benefits into short-term benefits, post-retirement benefits and other long term benefits based on the accrual and settlement of the benefits.

In case of in service health benefits:

- Any benefit that is settled within 12 months for the balance sheet date are classified as short-term benefits.
- Short term benefits are recognized on undiscounted basis without need for any complex assumptions.
- The benefits for active employees are annually renewed. The premiums paid cover the benefits receivable over next 12 months.
- The benefit for future years is not accrued or vested as on the current date.
- The company may recognize a short term asset corresponding to the prepaid premium.
- This does not require involvement of actuary as estimate of prepaid asset / unutilized premium is typically done on a straight line basis by the company.

In case of post retirement medical benefits:

- Any benefit that is payable after the exit of the employee is classified as post-retirement benefit.
- The post retirement benefit accrues over the period of vesting of the benefit.
- The obligation exists but not settled till the death of member. The company needs to reflect the same in its books of accounts.
- In this case the benefit accrues from the date of joining to the date of retirement.

- The company needs to estimate the total obligation at the point of retirement and provide for the accrued portion of this obligation as on the balance sheet date.
 - This requires use of assumptions like discounting rate, future medical premium, inflation rates, mortality rates, attrition rates etc.
 - The Indian accounting standard encourages involvement of an actuary to make these cost estimations using the Projected Unit Credit method.
- (6)
- ii. Following data points would be normally required for valuation of PRMS:
- Scheme rules
 - Previous valuation reports
 - Employee data
 - Identifier
 - DOB
 - DOJ
 - Membership status (active/ primary beneficiary / dependent beneficiary)
 - Gender
 - Whether spouse is alive
 - DOB of spouse
 - Unutilized domiciliary limit as on the date of valuation
 - Financial data
 - Premium paid during the year
 - Out patient benefits paid during the year.
 - Any exceptional benefits paid by the company
 - Asset reconciliation from opening to closing date.

Considerations while setting assumptions:

- Discount rate is based on the term of liability. The Indian government bonds yields corresponding to the term of obligation is taken as discount rate.
- Pre-retirement and post-retirement mortality assumption is based on the standard table. Adjustments to the table may be made in case the size of company is very large and sufficient past data is analyzed for mortality experience.
- Attrition assumption needed to project the chance of any employees retiring and thus eligible for benefit.
- Assumption for future premium applicable from the retirement of the employee is needed. The current average premiums would serve as starting point. However, the company can use any other reasonable cost estimated so long as it is backed by reasonable logic.
- Assumption for medical inflation in this assumed premium is needed.
- Age gap between member and spouse needs to be assumed if spouse's DOB is not available when the member is still alive.
- Need to assume an average domiciliary benefit consumed by the members as it could be possible that all the members do not consume the full 50,000 limit.
- Assumptions should be mutually consistent and best estimate of the expected actual experience.
- An analysis of past few years data may be needed to set some of the above assumptions.
- It must be noted that past experience may not be indicative of the future.

Additional data points required for valuation of PRMS:

- The premium cashflows after the retirement of employee is projected based on the assumed premium and premium inflation.
- The domiciliary benefit cashflows after the retirement of employee is projected based on the assumed consumption of the benefit and inflation on this benefit.
- The cashflows of domiciliary benefit need to be limited to the ceiling under the plan.
- The expected numbers are computed based on the assumed mortality.
- The dependents benefit is estimated using the expected benefit cashflows and probability of death of primary member for each cashflow year.
- Certain adjustments to the assumed cost may be needed as premium of single life would be different than premium for family.
- The present value is computed using the assumed discount rate.
- The value of obligation is the accrued portion of the above estimate based on the period of past service and expected total service of employee. This valuation methodology is known as Projected Unit Credit methodology.

(16)

iii. Considerations while revising assumptions:

- Need to confirm the reason for spike in claims (and hence the premiums)
- Need to verify if the increase in premiums is commensurate with the excess of actual claims over expected claims.
- Need to verify if there was any difference in the coverage for the year. Eg: current year's policy may cover the domiciliary benefit. Hence even if the premium is higher, the assumption for domiciliary benefit would reduce correspondingly and would have offsetting impact.
- Need to verify if there was an error in assuming the premiums in last valuation Eg. May be the amounts assumed were exclusive of taxes whereas current year number are including taxes.
- Is the increase in premium an industry wide event or whether other insurer can offer same cover for lower premium.
- No changes may be needed in the assumption in case of an exceptional event.
- The assumption for expected average premium may be revised if the insurer indicates that the premiums are unlikely to return to previous year's level.
- Assumption for inflation may need to be revised in case the actual premiums are growing at a rate higher than 8% for a considerable past period.
- Medical expenses increase in steps. It is possible that the premiums did not increase in last few years but increased steeply in current year.
- The need for change in assumption depends on the long term observed average increase.
- Overall inflation in the economy affecting the medical costs
- Changing demography of the employee population should also be considered, i.e. in case of death of a younger member during the year, average age of population would go up.

(6)

iv. Changes to various items of disclosures:

- The service cost and closing obligation will be revised as the premiums expected from the active employees when they retire is revised upwards.

- The extent of increase will only be for the hospitalization portion and is likely to be in proportion of the increase in assumed premium
 - The impact of change in premium in the obligation will be recognized in the actuarial loss due to change in financial assumptions.
 - In case the medical inflation rate changes, the impact will be approx. 20% of hospitalization portion of liability for each 100 bps change in assumption. This is assuming the duration of Post Retirement Medical plan is 20 years considering benefit is paid for the whole life.
 - The change will be recognized in the actuarial loss due to change in financial assumptions.
 - There will not be any impact on interest cost.
 - The difference between the expected premiums and actual premiums will be recognized in the gains/losses due to participant experience.
 - The participant experience would also arise due to deviation in other assumptions.
 - Experience loss due to actual increase in premium of 40% against the medical inflation assumption of 8% is expected to be ~30% of hospitalization liability.
- (6)

v. Derisking the plan

- Reduce benefits
 - Lower the sum assured
 - Lower domiciliary limit
 - Offer benefit only to primary member
 - Offer benefit only till age 75
 - Start the benefit from higher age, say 65

	Employer	employee
Pros	• Lower premiums for lower benefits • Improves long term feasibility of scheme	• Improves long term feasibility of scheme • Higher utilization of the benefit as some of the benefits may not be fully utilized by the members. • Chance to negotiate other alternate benefit (like retirement ex-gratia)
Cons	• Possible demotivation in workforce • The reduction in premiums may not be in same proportion as reduction in benefits	• The retired members dependent on the benefit may find it hard to get alternate service provider given the age.

- Increase contributions
 - Introduce subscription fees
 - Introduce sharing of premiums (company pays 75% employee pays 25%)

	Employer	Employee
Pros	• Reduced costs	• As the scheme has become cheaper, the employer will be

	Higher appreciation of the benefit by the members	- able to run the scheme for longer period. - The cost will be still cheaper than premium payable by the member should they buy a policy in open market
Cons	- Members may drop out thus making the scheme unattractive - Administrative overhead to collect and account for the receivable member contributions	- Not all members may have required cash to pay annually to the company. - Non-payment of premium may lead to lapse in benefits.

- Self administer the scheme
 - Discontinue insurance and pay the benefits from trust assets.

	Employer	Employee
Pros	- Reduced costs as there are no profit margins or other overheads - Freedom in choosing investments - Freedom for contributions. i.e. no sudden cash-flow in the form of annual premium	Need to deal with the company for benefits. No third party involvement.
Cons	- Need infrastructure similar to TPA to administer the benefits - Company may not have required expertise or risk appetite to deal with sudden events like epidemics or other healthcare trends - Company may not be able to detect false claims	- Company unlikely to have any cashless arrangements. The member will have to seek reimbursements after the treatment. Hence need to have access to liquid cash all the time. - The company is not regulated for these benefits (as against any health insurer) and hence the security of benefit may not be as high as that of an insurer.

(12)

- vi. Considerations while creating a contribution plan and the asset allocation for the fund over the next 5 years
- Need to check the growth plan of the company as regards the workforce.
 - The obligation for next 5 years needs to be worked out.
 - The obligations need to allow for:
 - Future entrants, retirees, deaths
 - Future growth in insurance premiums from the currently assumed values

- Any changes to the benefit levels after confirming the same with management.
- The company needs to specify if contribution plan should be based on level annual contributions or salary linked contributions or any other basis
- The amount of contribution or percentage can then be determined such that the accumulated value over 5 year period is equal to the projected obligation.
- The contributions will also depend on the assumed rate of return on assets. The rate of return in turn depends on the asset allocation.
- Frontloading contributions may help in building up the fund earlier but may lead to concerns in business cash-flows and overall financial well-being of the company.
- Account for contributions from employees / retirees as co-payment to medical premium in estimating employer's costs.
- Deferring the contributions to the later years may reduce the security of benefits as future management / circumstances may require the company to deviate from the targets.
- Considerations for setting up asset allocations:
 - Whether the outgo in the form of premiums is expected to be from the assets or from the company's funds in the initial years.
 - Fund needs to hold liquid assets to the extent that near term liabilities can be met easily.
 - Assets backing medium and long term outflows can be invested in riskier instruments like corporate bonds or equities.
 - Need to check rules governing limits under each asset category.
 - The expected rate of return on total fund will then be weighted average of expected returns on each asset class.
 - Asset allocation should also consider the taxation environment as any need to liquidate assets prematurely may trigger tax liabilities.
 - The asset allocation should reflect company's risk appetite towards these assets and liabilities.

(8)

[54 Marks]
